JACC: CARDIOVASCULAR INTERVENTIONS VOL. 18, NO. 1, 2025
© 2025 THE AUTHORS. PUBLISHED BY ELSEVIER ON BEHALF OF THE AMERICAN COLLEGE OF CARDIOLOGY FOUNDATION. THIS IS AN OPEN ACCESS ARTICLE UNDER THE CC BY-NC-ND LICENSE (http://creativecommons.org/licenses/by-nc-nd/4.0/).

ORIGINAL RESEARCH

STRUCTURAL

Surgical Bailout in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement

Incidence, Trends, and Clinical Outcomes

Hugo M. Aarts, MD,[a,b,*] Astrid C. van Nieuwkerk, MD,[a,*] Kimberley I. Hemelrijk, MD,[a] Jorge Salgado Fernandez, MD,[c] Didier Tchétché, MD,[d] Fabio S. de Brito, Jr, MD, PhD,[e] Marco Barbanti, MD,[f] Ran Kornowski, MD,[g] Azeem Latib, MD,[h,i] Augusto D'Onofrio, MD, PhD,[j] Flavio Ribichini, MD, PhD,[k] María Cruz Ferrer, MD,[l] Nicolas Dumonteil, MD,[d] Alexandre Abizaid, MD,[e] Samantha Sartori, PhD,[m] Paola D'Errigo, MSc,[n] Giuseppe Tarantini, MD, PhD,[j] Alberto Paolo Del Sole, MD,[k] Katia Orvin, MD,[g] Matteo Pagnesi, MD,[o] Eduardo Pinar, MD,[p] George Dangas, MD, PhD,[m] Roxana Mehran, MD,[m] Michiel Voskuil, MD, PhD,[b] Ronak Delewi, MD, PhD[a]

ABSTRACT

BACKGROUND Transcatheter aortic valve replacement (TAVR) harbors the risk of periprocedural complications that require emergent cardiac surgery, or "surgical bailout." Surgical bailout intends to be lifesaving but is associated with high mortality. This has given rise to discussion on the necessity of surgical backup during TAVR. Therefore, recent studies in contemporary TAVR patients are warranted.

OBJECTIVES This study aimed to assess the incidence, trends, and clinical outcomes of surgical bailout in patients undergoing transfemoral TAVR.

METHODS The CENTER2 (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves 2) study contains patient-level data from 10 clinical studies with patients undergoing TAVR between 2007 and 2022.

RESULTS A total of 24,010 patients undergoing transfemoral TAVR were included. The mean age was 81.5 ± 6.7 years, and 56% were female. The median Society of Thoracic Surgeons Predicted Risk of Mortality declined over time (2007-2014: 7.1% [Q1-Q3: 4.3%-14.8%] vs 2015-2022: 3.9% [Q1-Q3: 2.6%-6.0%]; $P < 0.001$). Surgical bailout was reported in 125 (0.52%) patients, and its incidence decreased over time, from 0.84% in 2007 to 2010 to 0.25% in 2019 to 2022 ($P_{trend} < 0.001$). Surgical bailout was more frequently observed in women compared with men (0.61% vs 0.41%; $P = 0.03$). Ventricular perforation was the most prevalent reason for surgical bailout (28.0%). Patients who underwent surgical bailout reported higher mortality rates during the 1-year follow-up compared with patients without surgical bailout (68.3% vs 15.3%; HR: 8.60; 95% CI: 6.74-10.97; $P < 0.001$). Interestingly, patients surviving the first 90 days reported a similar mortality risk (HR: 2.19; 95% CI: 0.91-5.27; $P = 0.08$).

CONCLUSIONS The incidence of surgical bailout is very low in the contemporary TAVR population. Mortality rates remain high among patients requiring surgical bailout during transfemoral TAVR, but this risk diminishes in patients surviving the postprocedural period. (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves 2 [CENTER2]; NCT03588247) (JACC Cardiovasc Interv. 2025;18:89-99) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

ISSN 1936-8798 https://doi.org/10.1016/j.jcin.2024.09.050

JACC: CARDIOVASCULAR INTERVENTIONS VOL. 18, NO. 1, 2025
JANUARY 13, 2025:89–99

ABBREVIATIONS AND ACRONYMS

TAVR = transcatheter aortic valve replacement

VARC-2 = Valve Academic Research Consortium-2

Transcatheter aortic valve replacement (TAVR) has emerged as a safe and minimally invasive treatment for patients with severe aortic valve stenosis. Recent developments in preprocedural planning, procedural techniques, and increased operator experience have improved procedural safety, leading to the expansion of indications for TAVR toward patients with lower surgical risk profiles.[1] However, TAVR procedures still harbor the risk of major periprocedural complications that require emergent cardiac surgery, or "surgical bailout". Surgical bailout intends to be lifesaving, but morbidity and mortality are very high.[2,3] Interestingly, recent studies on the incidence and clinical outcomes of surgical bailout in the contemporary TAVR population are scarce. An update on surgical bailout is warranted with the ongoing discussion on the implementation of TAVR in low-risk patients and the necessity of surgical backup in TAVR centers.[4] Therefore, the aim of this study was to assess the incidence, temporal trends, and clinical outcomes of surgical bailout in patients undergoing transfemoral TAVR.

METHODS

The CENTER2 (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves 2) study is an international collaboration including patients with severe aortic valve stenosis undergoing TAVR. The initial CENTER (Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves) study included 12,381 patients who underwent transfemoral TAVR between 2007 and 2018. This patient-level database consists of 1 prospective clinical trial, 2 multicenter prospective registries, 3 national registries, and 4 single-center prospective registries, which were identified through a systematic search. In April 2022, all collaborators were asked to add more contemporary patients to the database, which led to the addition of 13,390 contemporary patients to the new CENTER2 database. The design of the CENTER and CENTER2 studies has been described previously.[5,6]

The CENTER2 study includes a worldwide population because patients underwent TAVR in Spain, Italy, France, Israel, Brazil, and the United States. The collaborating centers provided a dedicated database including patient characteristics, echocardiographic measurements, procedural data, and long-term follow-up. Surgical risk profiles were estimated using the Society of Thoracic Surgeons Predicted Risk of Mortality and were divided into low risk (<4%), intermediate risk (4%-8%), and high risk (>8%). Indications for TAVR and the selection of the transcatheter heart valve and size were determined by the multidisciplinary heart team of each participating center. All individual studies were conducted according to the Declaration of Helsinki. Institutional Review Board approval was obtained at each center. All patients provided written informed consent at their local hospital. The study is registered at ClinicalTrials.gov (NCT03588247).

PATIENT POPULATION. The CENTER2 study consists of 25,771 patients undergoing TAVR. Transfemoral TAVR was performed in 24,321 (94.4%) patients. We included patients undergoing transfemoral TAVR in the current study. The incidence of surgical bailout was available in 24,010 (98.7%) patients undergoing transfemoral TAVR. All TAVR procedures were performed in heart centers with on-site surgical backup.

STUDY DEFINITIONS AND OUTCOMES. The primary outcome of the current study was the incidence of emergent cardiac surgery, or surgical bailout, in

From the [a]Department of Cardiology, Amsterdam University Medical Center, University of Amsterdam, Amsterdam Cardiovascular Sciences, Amsterdam, the Netherlands; [b]Department of Cardiology, University Medical Center Utrecht, Utrecht, the Netherlands; [c]Complejo Hospitalario Universitario, A Coruña, Spain; [d]Clinique Pasteur, Toulouse, France; [e]Heart Institute, University of São Paulo Medical School, São Paulo, Brazil; [f]Università degli Studi di Enna "Kore," Enna, Italy; [g]Cardiology Department, Rabin Medical Center, Petah Tikva, Israel; [h]Department of Cardiology, Montefiore Medical Center, New York, New York, USA; [i]Division of Cardiology, Department of Medicine, University of Cape Town, Cape Town, South Africa; [j]Division of Cardiac Surgery, University of Padova, Padova, Italy; [k]Division of Cardiology, Department of Medicine, University of Verona, Verona, Italy; [l]Servicio de Cardiología, Hospital Miguel Servet, Zaragoza, Spain; [m]The Zena and Michael A. Wiener Cardiovascular Institute, Icahn School of Medicine at Mount Sinai, New York, New York, USA; [n]National Centre for Global Health, Istituto Superiore di Sanità, Rome, Italy; [o]Institute of Cardiology, Azienda Socio Sanitaria Territoriale Spedali Civili, Department of Medical and Surgical Specialties, Radiological Sciences and Public Health, University of Brescia, Brescia, Italy; and the [p]Hospital Clínico Universitario Virgen de la Arrixaca, Murcia, Spain. *Drs Aarts and Nieuwkerk contributed equally to this work and are co-first authors.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received May 16, 2024; revised manuscript received September 5, 2024, accepted September 17, 2024.

TABLE 1 Baseline Patient and Procedural Characteristics in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement

	Total Cohort (N = 24,010)	Surgical Bailout (n = 125)	No Surgical Bailout (n = 23,885)	P Value
Demographics				
Age, y	81.5 ± 6.7	82.4 ± 5.8	81.4 ± 6.7	0.11
Women	13,415 (56)	82 (66)	13,333 (56)	0.03
Body mass index, kg/m^2	27.4 ± 4.9	27.1 ± 5.8	27.4 ± 4.9	0.46
Medical history				
Coronary artery disease	8,693 (38)	37 (31)	8,656 (38)	0.10
Prior myocardial infarction	2,958 (13)	9 (7)	2,949 (13)	0.07
Prior PCI	4,797 (21)	18 (15)	4,779 (21)	0.11
Prior CABG	2,045 (9)	6 (5)	2,039 (9)	0.14
Dyslipidemia	13,825 (59)	71 (58)	13,754 (59)	0.91
Diabetes mellitus	7,697 (33)	27 (22)	7,697 (33)	0.01
Hypertension	19,184 (81)	98 (80)	19,086 (81)	0.82
Peripheral vascular disease	2,824 (12)	4 (3)	2,824 (12)	0.003
Atrial fibrillation	6,499 (28)	36 (31)	6,463 (28)	0.58
Previous cerebrovascular events	2,395 (10)	14 (12)	2,381 (10)	0.62
Renal failure	2,587 (12)	9 (8)	2,578 (12)	0.13
NYHA functional class ≥3	8,765 (51)	46 (55)	8,718 (51)	0.37
Risk scores				
Logistic EuroSCORE, %	13.5 (8.4-21.4)	13.6 (9.0-19.9)	13.5 (8.4-21.4)	0.70
EuroSCORE II, %	3.6 (2.2-6.0)	3.0 (2.3-5.4)	3.7 (2.2-6.0)	0.17
STS-PROM, %	4.9 (3.1-8.5)	5.0 (3.5-9.5)	4.9 (3.1-8.5)	0.41
Echocardiographic parameters				
LVEF, %	56.9 ± 13.1	60.2 ± 13.3	56.8 ± 13.1	0.03
Aortic mean gradient, mm Hg	48.9 ± 16.5	52.3 ± 17.8	48.9 ± 16.5	0.03
Aortic maximum gradient, mm Hg	77.7 ± 23.2	81.1 ± 26.2	77.7 ± 23.2	0.14
Aortic valve area, cm^2	0.67 ± 0.20	0.65 ± 0.22	0.67 ± 0.20	0.35
Procedural characteristics				
Balloon-expandable valve	10,377 (43)	70 (57)	10,307 (43)	0.003
Newer-generation valves	13,606 (57)	45 (37)	13,561 (57)	<0.001
Valve in valve	590 (3)	4 (4)	586 (3)	0.54
Predilatation	8,411 (51)	53 (60)	8,358 (51)	0.07
Postdilatation	4,306 (22)	20 (19)	4,286 (22)	0.53
Valve size, mm	26 (25-29)	26 (23-29)	26 (25-29)	0.84

Values are mean ± SD, n (%), or median (Q1-Q3).

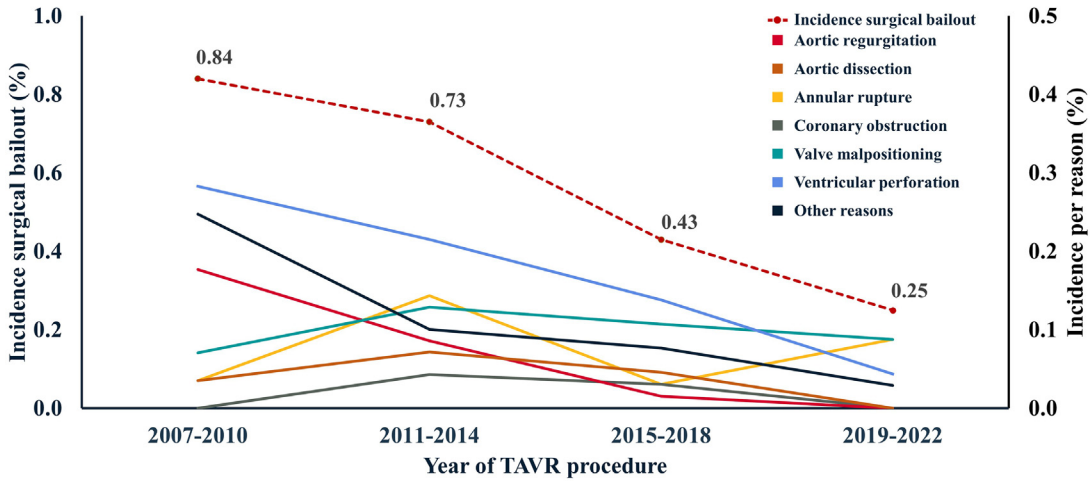
CABG = coronary artery bypass graft; LVEF = left ventricular ejection fraction; PCI = percutaneous coronary intervention; STS-PROM = Society of Thoracic Surgeon Predicted Risk of Mortality; TAVR = transcatheter aortic valve replacement.

patients undergoing transfemoral TAVR. Surgical bailout was defined in accordance with the Valve Academic Research Consortium-2 (VARC-2).[7] The VARC-2 defines surgical bailout as conversion to open sternotomy secondary to any procedure-related complications. The secondary outcomes of the current study were clinical outcomes of patients undergoing surgical bailout. Clinical outcomes were defined in accordance with the VARC-2. All participating centers were asked to report the primary reason and timing of surgical bailout. Transcatheter heart valves were considered as newer generation if they became commercially available after 2013. Supplemental Table 1 provides an overview of the transcatheter heart valves used.

STATISTICAL ANALYSIS. Continuous variables were visually tested for normality with distribution plots. Accordingly, variables were reported as mean ± SD or median ± IQR), and differences were tested with the independent *t*-test or the Mann-Whitney *U* test. Categoric variables were reported as frequencies and percentages. Differences between groups were tested with the chi-square test or the Fisher exact test as appropriate. The incidence of surgical bailout was assessed, and changes in the incidence of surgical bailout over time were tested with the chi-square Mantel-Haenszel test for trend. Differences in the incidence of surgical bailout for different subgroups were tested with the chi-square test, and ORs with 95% CIs were established using logistic regression. Subgroup analysis was performed for sex, surgical risk profiles, and balloon- and self-expandable valves.

Differences in clinical outcomes in patients with and without surgical bailout were tested with the chi-square test, and ORs with 95% CIs were established using logistic regression. Time-to-mortality estimates

FIGURE 1 Temporal Trends in the Incidence and Reasons for Surgical Bailout in Patients Undergoing Transfemoral TAVR



TAVR = transcatheter aortic valve replacement.

were made with Cox regression, and HRs were reported. To assess the effect of surgical bailout on mortality after the initial postprocedural period, Cox regression analysis with the landmark time at 90 days was performed. The aim of this Cox regression analysis was to assess the effect of surgical bailout on mortality between 90 and 365 days. The reference category for ORs and HRs consisted of patients without surgical bailout. Subgroup analyses were performed for time periods and reasons for surgical bailout. All statistical tests were 2-tailed, and a P value <0.05 was considered statistically significant. Calculations were generated by SPSS software (version 28.0 for Windows, IBM Corp).

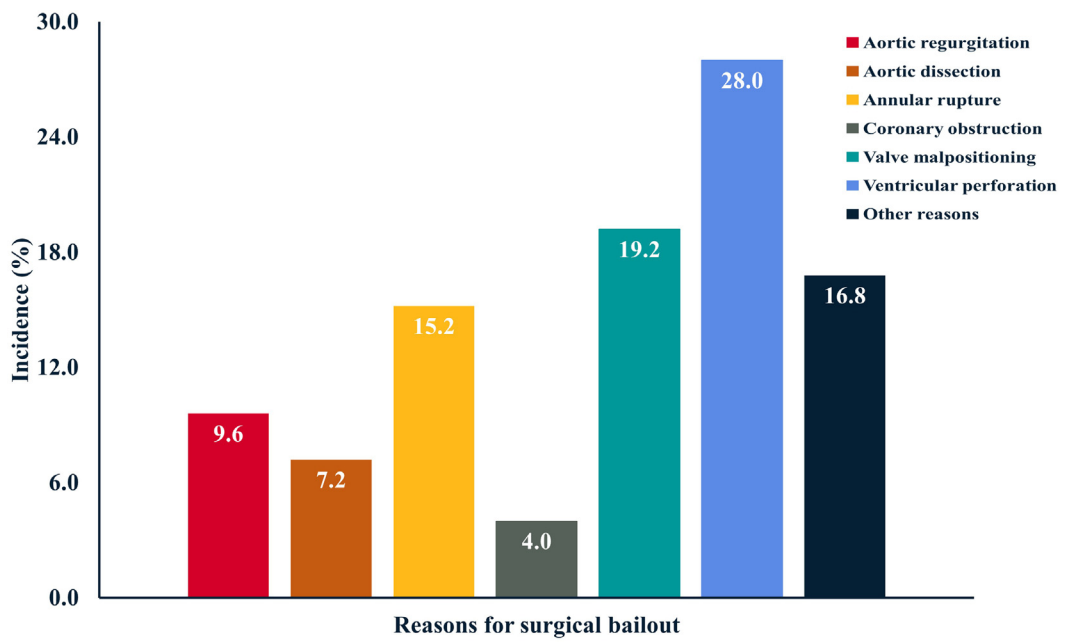
RESULTS

A total of 24,010 patients who underwent transfemoral TAVR were included in the current study. The mean age was 81.5 ± 6.7 years, and 56% were female. The median Society of Thoracic Surgeons Predicted Risk of Mortality was 4.9% (Q1-Q3: 3.1%-8.5%), and it declined over time (2007-2014: 7.1% [Q1-Q3: 4.3%-14.8%] vs 2015-2022: 3.9% [Q1-Q3: 2.6%-6.0%]; P < 0.001). Newer-generation valves were used in 57% of patients, and 43% of patients were treated with balloon-expandable valves. The median follow-up was 364 days (Q1-Q3: 40-654 days). **Table 1** presents baseline characteristics of the included cohort.

INCIDENCE OF SURGICAL BAILOUT. A total of 125 (0.52%) patients undergoing transfemoral TAVR underwent surgical bailout, with 91.0% of these cases taking place immediately after the TAVR procedure. The incidence of surgical bailout decreased over time (0.84% in 2007-2010, 0.73% in 2011-2014, 0.43% in 2015-2018, and 0.25% in 2019-2022; P_{trend} < 0.001; **Figure 1**). **Table 1** displays the baseline characteristics of patients who did and did not undergo surgical bailout. Surgical bailout was more frequently observed in women compared with men (0.61% vs 0.41%; OR: 1.51; 95% CI: 1.04-2.18; P = 0.03). Patients with low surgical risk profiles reported similar rates of surgical bailout compared with patients with intermediate and high surgical risk profiles (low risk: 0.48% vs intermediate risk: 0.62% vs high risk: 0.54%; P = 0.65). Patients treated with balloon-expandable valves had higher rates of surgical bailout compared with patients undergoing transfemoral TAVR with self-expandable valves (0.67% vs 0.40%; OR: 1.70; 95% CI: 1.19-2.43; P = 0.003).

Figure 2 displays the various reasons for surgical bailout. Ventricular perforation was the most prevalent reason for surgical bailout (28.0%), followed by valve malpositioning (19.2%) and annular rupture (15.2%). Ventricular perforation causing surgical bailout occurred primarily in the left ventricle (84.6%), mostly caused by the stiff guidewire. Pacing over the stiff guidewire was not used in any of these cases. Moreover, ventricular perforation was the main reason for surgical bailout during the largest time frame of the study, but valve malpositioning and annular rupture were the most frequently reported reasons for surgical bailout in the last 4 years of the

FIGURE 2 Reasons for Surgical Bailout in Patients Undergoing Transfemoral Transcatheter Aortic Valve Replacement

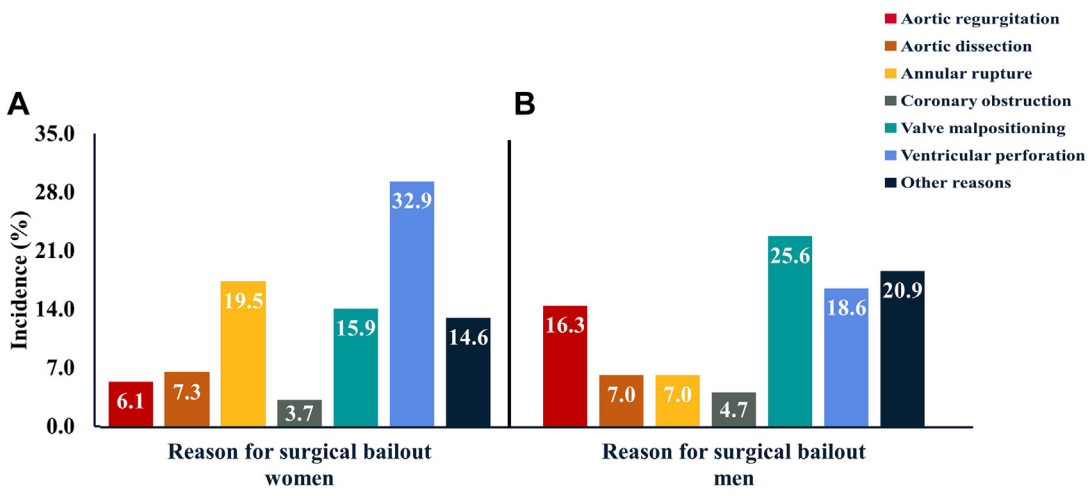


current study (Figure 1). Severe aortic regurgitation was more frequently reported among patients who underwent surgical bailout during the hospital stay compared with patients who underwent surgical bailout immediately after transfemoral TAVR (42.9% vs 2.2%; $P = 0.002$). The reasons for surgical bailout differed for women and men ($P = 0.049$) (Figure 3), with annular rupture (19.5% vs 7.0%; $P = 0.02$) and ventricular perforation (32.9% vs 18.6%; $P = 0.01$) more frequently reported as the reason for surgical bailout among women compared with men. Patients treated with balloon- and self-expandable valves reported different reasons for surgical bailout after transfemoral TAVR ($P = 0.006$) (Figure 4). Annular rupture was the predominant reason for surgical bailout in patients treated with balloon-expandable valves and occurred more frequently in these patients compared with those treated with self-expandable valves (24.3% vs 3.7%; $P < 0.001$). Ventricular perforation (38.9%) was the most important reason for surgical bailout in patients undergoing TAVR with self-expandable valves.

CLINICAL OUTCOMES IN PATIENTS REQUIRING SURGICAL BAILOUT. Table 2 presents the clinical outcomes in patients with and without surgical bailout during transfemoral TAVR. A total of 93 (74.4%) patients survived the emergent surgical intervention initially. Procedural mortality was more frequently observed in patients who underwent surgical bailout after transfemoral TAVR compared with patients without surgical bailout (25.6% vs 1.1%; OR: 31.50; 95% CI: 20.69-47.93; $P < 0.001$). Accordingly, in-hospital mortality was also higher in patients with surgical bailout compared with patients who did not undergo surgical bailout (48.0% vs 3.5%; OR: 25.71; 95% CI: 17.97-36.96; $P < 0.001$). Mortality risk in patients with surgical bailout was higher during the first 30 days (51.7% vs 4.7%; OR: 21.49; 95% CI: 14.90-31.02; $P < 0.001$) and during the 1-year follow-up after transfemoral TAVR compared with patients who did not undergo surgical bailout (68.3% vs 15.3%; HR: 8.60; 95% CI: 6.74-10.97; $P < 0.001$) (Figure 5). However, patients who underwent surgical bailout were no longer at increased risk if they survived the first 90 days after transfemoral TAVR (HR: 2.19; 95% CI: 0.91-5.27; $P = 0.08$) (Figure 6). In addition, follow-up beyond 1 year was available in 31 (55.4%) patients who survived the first year after surgical bailout (median follow-up of 731 days [Q1-Q3: 402-1,611 days]). A total of 9 (29.0%) patients died during follow-up, with a median time of 981 days (Q1-Q3: 683-2,050) between the TAVR procedure and death.

Thirty-day mortality in patients who underwent surgical bailout did not decrease in contemporary patients undergoing transfemoral TAVR between 2015 and 2022 compared with those treated between 2007 and 2014 (2007-2014: 51.3% vs 2015-2022: 52.4%; $P = 0.91$). Similarly, patients undergoing transfemoral TAVR during the coronavirus disease-2019 pandemic

FIGURE 3 Reasons for Surgical Bailout in Women and Men Undergoing Transfemoral Transcatheter Aortic Valve Replacement



Reasons for surgical bailout in (A) women and (B) men undergoing transfemoral transcatheter aortic valve replacement.

were found to have comparable mortality rates as patients treated between 2017 and 2019 (2017-2019: 57.9% vs 2020-2022: 45.5%; $P = 0.51$). Mortality rates during the first 30 days differed among the various reasons for surgical bailout ($P = 0.025$) (Figure 7). Patients who underwent surgical bailout because of annular rupture were found to have the highest 30-day mortality (68.4%), followed by patients with ventricular perforation (66.7%).

Patients who were discharged after surgical bailout reported a median hospital stay of 14 days (Q1-Q3: 9-29 days) compared with a median of 6 days (Q1-Q3: 4-9 days) in patients without surgical bailout ($P < 0.001$). In patients who underwent surgical bailout after transfemoral TAVR, there were also higher rates of major bleeding (38.0% vs 6.4%; OR: 8.96; 95% CI: 6.19-12.99; $P < 0.001$) and myocardial infarction (7.5% vs 1.0%; OR: 7.68; 95% CI: 3.69-16.00; $P < 0.001$) at the 30-day follow-up. The incidence of 30-day stroke did not differ between patients with and without surgical bailout (4.0% vs 2.2%; OR: 1.85; 95% CI: 0.75-4.55; $P = 0.17$) and neither did the incidence of disabling stroke affecting daily functional status (1.6% vs 0.7%; OR: 2.44; 95% CI: 0.60-9.97; $P = 0.20$).

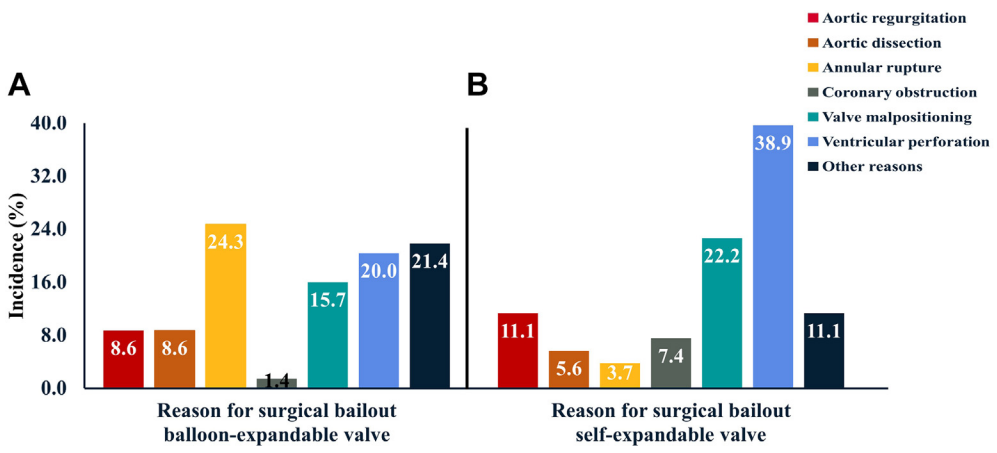
DISCUSSION

The key findings of the current study on surgical bailout in this contemporary TAVR cohort including more than 24,000 patients are as follows: 1) surgical bailout was reported in 0.52% of patients undergoing transfemoral TAVR; 2) the incidence of surgical bailout declined over time from 0.84% in 2007 to 2010 to 0.25% in 2019 to 2022; 3) women reported higher rates of surgical bailout after transfemoral TAVR; 4) ventricular perforation was the most prevalent reason for surgical bailout; and 5) surgical bailout was associated with very high mortality risk, which diminished after the initial postprocedural period (Central Illustration).

TAVR procedures still carry the risk of major periprocedural complications that require conversion to open heart surgery, so-called surgical bailout. Results from the STS-ACC TVT (Society of Thoracic Surgeons-American College of Cardiology Transcatheter Valve Therapy) Registry found that surgical bailout was performed in 1.17% of patients undergoing TAVR between 2011 and 2015 in the United States.[2] Another multicenter registry found that surgical bailout was required in 0.76% of patients undergoing transfemoral TAVR between 2013 and 2016.[3] We found a much lower incidence of surgical bailout among patients undergoing transfemoral TAVR between 2007 and 2022. The results from our worldwide TAVR cohort also indicate that the incidence of surgical bailout declined over the last decade; only 0.25% of patients underwent surgical bailout after transfemoral TAVR in recent years.

Surprisingly, we did not observe lower rates of surgical bailout among low-risk patients compared with patients with increased risk during conventional surgery. This finding suggests that the need for surgical bailout may be predominantly procedure

FIGURE 4 Reasons for Surgical Bailout in Patients Treated With Balloon-Expandable Valves and Self-Expandable Valves



Reasons for surgical bailout in patients treated with (A) balloon-expandable valves and (B) self-expandable valves.

related. It emphasizes that risk assessment based on patient characteristics before TAVR procedures is challenging. This finding should be taken into account because TAVR indications are rapidly expanding toward patients with low surgical risk profiles. Particularly in low-risk patients who are considered both eligible for TAVR and conventional surgical aortic valve replacement, the risk of severe periprocedural complications requiring surgical bailout should be discussed carefully. On the other hand, the declining incidence of surgical bailout during the study period underscores that TAVR procedures have become much safer over the last decade, which can be ascribed to several factors including better preprocedural assessment by computed tomography angiography, more experienced operators, and newer-generation valves accompanied by smaller delivery systems.

REASONS FOR SURGICAL BAILOUT. In line with the aforementioned registries, ventricular perforation was the most prevalent reason for surgical bailout in this contemporary TAVR cohort. The significant proportion of patients requiring surgical bailout because of ventricular perforation suggests that TAVR procedures may benefit from the implementation of a more minimalistic approach. This minimalistic TAVR approach includes pacing over the stiff guidewire, which is routinely placed in the left ventricle, eliminating the need for transvenous temporary pacing in the right ventricle. The latter was shown to be the primary reason in 6.6% of surgical bailout procedures in a European TAVR cohort.[3] Our study observed fewer patients with right-sided ventricular perforations, even without the routine use of pacing over the guidewire. Therefore, further simplification of TAVR procedures may not only improve procedural duration but also further increase TAVR safety. Importantly, more data on the safety and effectiveness of this minimalistic approach are warranted.

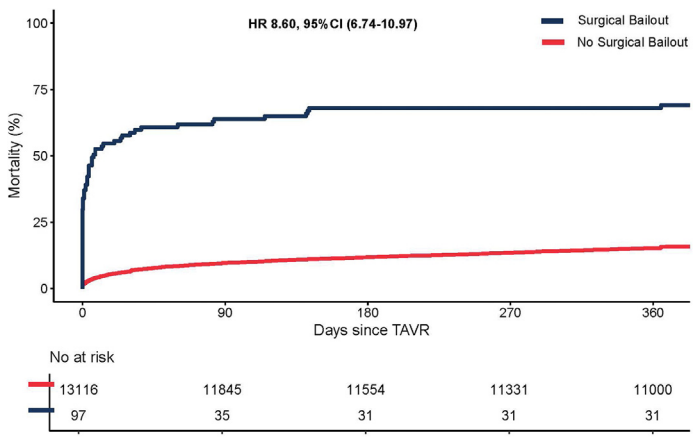
TABLE 2 Clinical Outcomes in Patients With and Without Surgical Bailout During Transfemoral Transcatheter Aortic Valve Replacement

	Surgical Bailout (n = 125)	No Surgical Bailout (n = 23,885)	OR (95% CI)
Procedural outcomes			
Mortality ≤72 h	32 (25.6)	258 (1.1)	31.50 (20.69-47.93)
30-day outcomes			
Mortality	61 (51.7)	975 (4.7)	21.49 (14.90-31.02)
Cardiovascular mortality	60 (50.8)	814 (4.1)	24.07 (16.63-34.76)
Noncardiovascular mortality	1 (1.7)	129 (0.7)	2.62 (0.36-19.09)
Stroke	5 (4.0)	525 (2.2)	1.85 (0.75-4.55)
Disabling stroke	2 (1.6)	160 (0.7)	2.44 (0.60-9.97)
Major bleeding	46 (38.0)	1,490 (6.4)	8.96 (6.19-12.99)
Myocardial infarction	8 (7.5)	197 (1.0)	7.68 (3.69-16.00)
Major vascular complication	9 (11.3)	129 (0.9)	14.76 (7.22-30.16)
Minor vascular complication	21 (26.3)	1,503 (9.6)	3.35 (2.03-5.54)
Permanent pacemaker	10 (8.0)	3,483 (15.1)	0.49 (0.26-0.94)
New-onset atrial fibrillation	7 (9.5)	706 (5.9)	1.69 (0.77-3.68)
Length of hospital stay, d	8 (3-19)	6 (4-9)	–
Days in intensive care unit	4 (1-8)	2 (1-3)	–
1-y outcomes			
Mortality[a]	69 (68.3)	2,116 (15.3)	8.60 (6.74-10.97)
Stroke[a]	7 (7.6)	719 (6.9)	1.07 (0.44-2.57)

Values are n (%). 30-day follow-up was complete in 20,675 (86.1%) of patients. One-year follow-up was complete in 13,966 (58.2%) of patients. [a]HR is reported for 1-year mortality and 1-year stroke.

JACC: CARDIOVASCULAR INTERVENTIONS VOL. 18, NO. 1, 2025
JANUARY 13, 2025:89–99

FIGURE 5 Time to Mortality Curves for Patients With and Without Surgical Bailout During Transfemoral TAVR

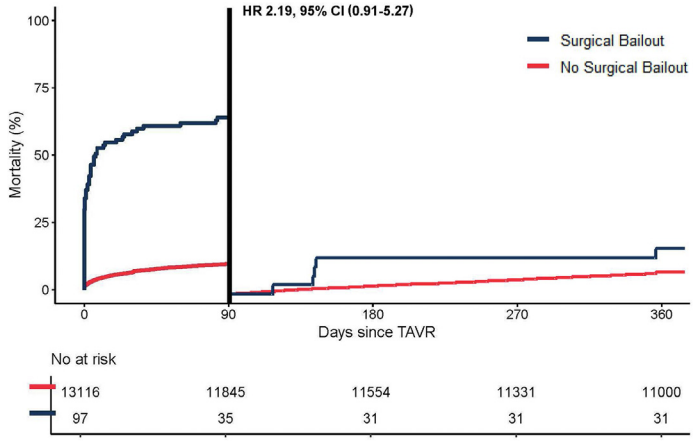


TAVR = transcatheter aortic valve replacement.

Notably, the number of patients undergoing surgical bailout because of annular rupture was substantial in the current study. Annular rupture was the most prevalent reason for surgical bailout among patients treated with balloon-expandable valves; annular rupture was 9-fold more frequently observed in these patients compared with patients undergoing TAVR with self-expandable valves. It remains 1 of the most feared periprocedural complications and has been shown to be associated with the use of balloon-expandable valves in the presence of severe calcifications in the left ventricular outflow tract.[8] Thus, it was somewhat remarkable that we did not observe a decline in annular rupture despite refinements in preprocedural assessment of the aortic root and increased operator experience with subsequent better procedural outcomes.[9] Additionally, we observed an increased risk of annular rupture and ventricular perforation in women undergoing transfemoral TAVR. Women with severe aortic stenosis undergoing TAVR are often characterized by small aortic annuli and smaller left ventricular dimensions compared with men. Our findings again emphasize the importance of randomized controlled trials investigating the optimal treatment in women undergoing TAVR. Moreover, the higher incidence of ventricular perforations among patients treated with self-expandable valves may be ascribed to more wire manipulation during valve deployment in these patients.

More encouraging was the observation that all other reasons for surgical bailout decreased over time, including severe aortic regurgitation and aortic dissection. Certainly, the formation of dedicated multidisciplinary TAVR teams including cardiologists, cardiac surgeons, and radiologists significant improved (pre)procedural techniques have minimized the risk of numerous feared periprocedural complications with subsequent need for surgical bailout.

FIGURE 6 Time to Mortality Curve With Landmark at 90 Days for Patients With and Without Surgical Bailout During Transfemoral TAVR

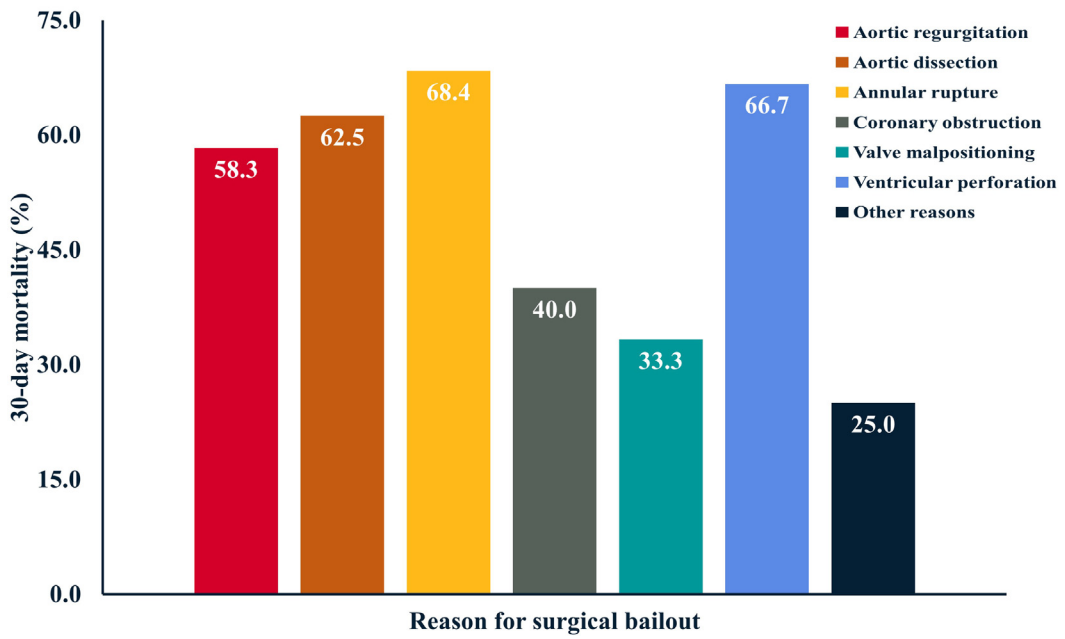


TAVR = transcatheter aortic valve replacement.

CLINICAL OUTCOMES AFTER SURGICAL BAILOUT. Our results confirm the dismal outcomes of patients undergoing surgical bailout at both short- and long-term follow-up. Previous studies suggested mortality rates ranging between 31.3% and 78.2% at 1-year follow-up.[2,3,10,11] In the current study, 1 in 4 patients who underwent surgical bailout died within 72 hours of the index procedure, whereas 1 in 2 patients did not survive the first 30 days after surgical bailout, which did not decline over the study period. Interestingly, we found that patients who survived the initial postprocedural period reported similar mortality rates at 1-year follow-up. This finding is hopeful and emphasizes that some patients indeed benefit from surgical bailout. It stresses the importance of on-site cardiac surgery teams in TAVR centers.

FUTURE PERSPECTIVES. Current international guidelines advocate that TAVR should be solely performed in heart centers with on-site surgical backup.[12] The improved safety of TAVR procedures has led to some discussion on the necessity of on-site surgical backup in TAVR centers. The presence of an on-site cardiac surgery team provides multiple advantages including

FIGURE 7 30-Day Mortality Rate per Reason for Surgical Bailout During Transfemoral TAVR

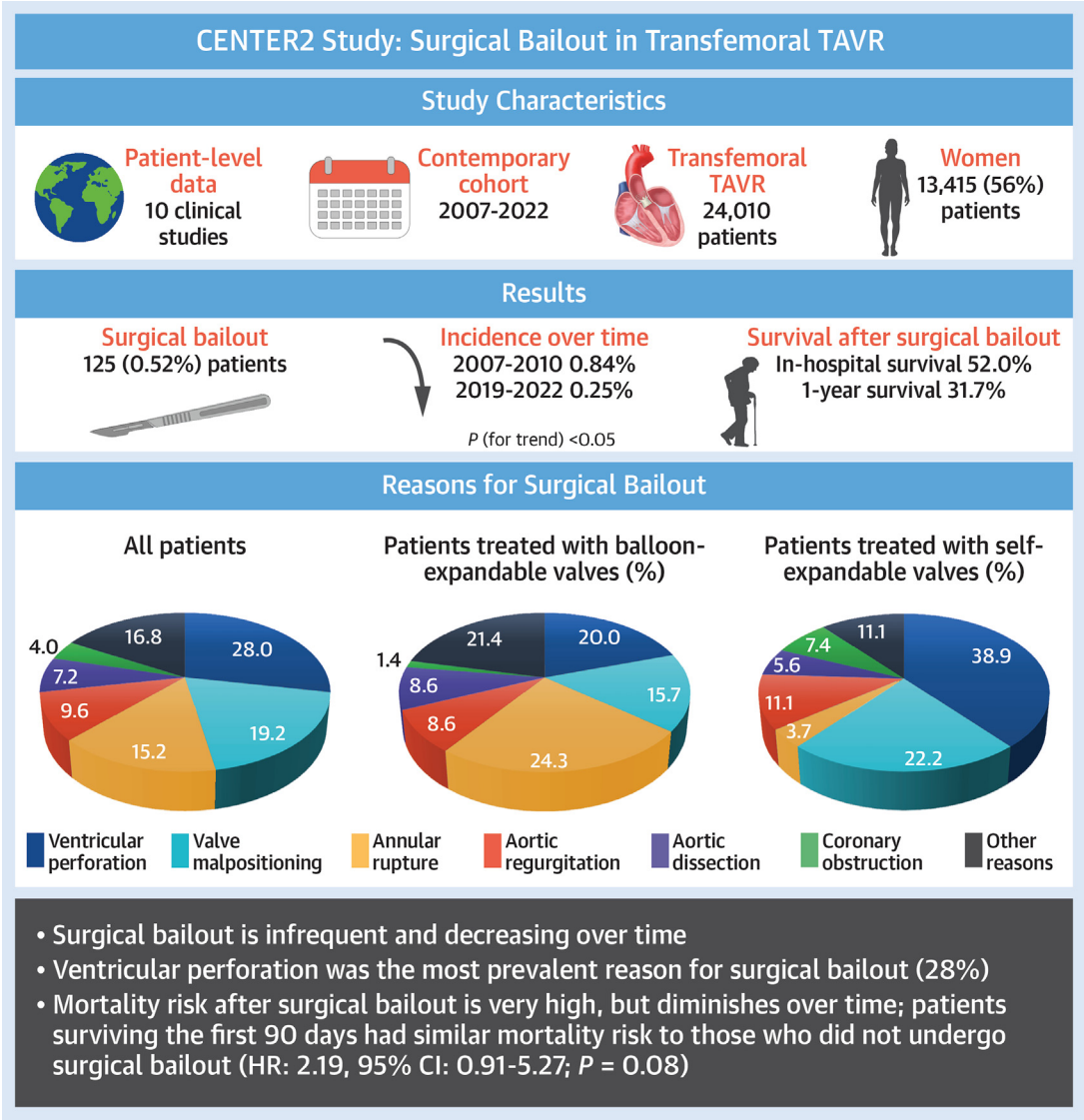


TAVR = transcatheter aortic valve replacement.

the important and valuable input from cardiac surgeons during heart team meetings and complex TAVR procedures. Moreover, clinical outcomes of patients undergoing TAVR have been shown to be better in heart centers with a high annual number of surgical aortic valve replacement procedures.[13] The current debate on the need for on-site surgical backup is primarily driven by the prolonged waiting time for TAVR procedures, which is associated with increased mortality in patients planned for TAVR.[14] The implementation of TAVR in heart centers without on-site surgical backup may improve accessibility and decreases waiting time for fragile patients. A recent study strengthened the belief that TAVR procedures can be safely performed in heart centers without on-site cardiac surgery because similar short-term outcomes after TAVR in heart centers with and without on-site cardiac surgery were reported.[4] However, these findings should be considered in light of some methodological limitations, including the retrospective design of the study with the potential of selection bias. Most importantly, the need for emergent conversion to open heart surgery is difficult to predict and primarily procedure related, and prompt intervention by experienced cardiac surgery teams is key and can be lifesaving. Our results confirm that surgical bailout may be lifesaving in cases of major periprocedural complications. Therefore, intuitively, most patients will favor heart centers with on-site surgical backup during their TAVR procedure despite the increased waiting time. Efforts should be made to streamline the work-up and hospital admissions in patients planned to undergo TAVR to reduce the waiting time, rather than opening low-volume heart centers without on-site surgical backup to reduce the waiting time.[13] This includes the omission of standard percutaneous coronary intervention in patients undergoing TAVR and early discharge.[15,16]

STUDY LIMITATIONS. Our study of a contemporary TAVR cohort including more than 20,000 patients undergoing transfemoral TAVR has several limitations. First of all, the observational design of our study has well-known inherent limitations. Second, clinical outcomes were site reported, and there was no central event adjudication. Moreover, detailed information on patients who required surgical bailout because of major periprocedural complications but were deemed inoperable was not reported by the participating centers. The registration of symptomatic hypoxic injury attributable to hypotension or hypoxia caused by major periprocedural complications may have been a valuable addition to the registration of other neurologic complications as recommended in the recently updated guidelines by the Valve Academic Research Consortium. Furthermore, patient-related outcomes such as quality of life

CENTRAL ILLUSTRATION CENTER2 Study: Surgical Bailout in Transfemoral Transcatheter Aortic Valve Replacement



Aarts HM, et al. JACC Cardiovasc Interv. 2025;18(1):89–99.

*P (for trend) <0.05. CENTER2 = Cerebrovascular Events in Patients Undergoing Transcatheter Aortic Valve Implantation With Balloon-Expandable Valves Versus Self-Expandable Valves; TAVR = transcatheter aortic valve replacement.

and functional and neurologic status were not registered, but may offer a more comprehensive understanding of patient well-being after surgical bailout. Lastly, the results of this study are from high-volume heart centers with on-site cardiac surgery and may not be generalized directly toward heart centers with lower annual numbers or without on-site cardiac surgical teams.

CONCLUSIONS

The incidence of surgical bailout is low in patients undergoing transfemoral TAVR; only 125 (0.52%) patients underwent surgical bailout. Its incidence decreased between 2007 and 2022. Women reported higher rates of surgical bailout. Ventricular perforation and annular rupture were among the most

prevalent reasons for surgical bailout. Annular rupture was more frequently reported among women and in patients treated with balloon-expandable valves, whereas ventricular perforations were more commonly observed in those undergoing transfemoral TAVR with self-expandable valves. Importantly, mortality rates during short-term follow-up remain extremely high among patients undergoing surgical bailout, but this risk diminishes in patients surviving the initial postprocedural period.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This research was supported by the Netherlands CardioVascular Research Initiative, the Dutch Heart Foundation (CVON 2018-28 and 2012-06 Heart Brain Connection), the Dutch Federation of University Medical Centres, the Netherlands Organisation for Health Research and Development, the Royal Netherlands Academy of Sciences, and the Netherlands Organization for Health Research and Development. Dr de Brito J_R is a proctor for Edwards Lifesciences and Medtronic. Dr Barbanti is a consultant for Edwards Lifesciences; and has received speaker honoraria from Medtronic and Biotronik. Dr Latib is a consultant for Medtronic; and has received honoraria from Abbott Vascular. Dr Pagnesi has received personal fees from Abbott Vascular. Dr Delewi has received educational grants from Boston Scientific, Edwards Lifesciences, and Abiomed. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Ronak Delewi, Heart Center, Amsterdam UMC, Meibergdreef 9, 1105AZ Amsterdam, the Netherlands. E-mail: r.delewi@amsterdamumc.nl.

PERSPECTIVES

WHAT IS KNOWN? TAVR has become a safe treatment modality for patients with severe aortic stenosis but still harbors the risk of major periprocedural complications that require emergent cardiac surgery, or "surgical bailout."

WHAT IS NEW? The incidence of surgical bailout is very low and decreased over time. Ventricular perforation was the most prevalent reason for surgical bailout. Surgical bailout is indeed lifesaving in some patients with major periprocedural complications, but mortality rates are very high.

WHAT IS NEXT? On-site cardiac surgery will remain crucial in TAVR centers. However, future studies should strive to further improve TAVR procedures, preventing the need for surgical bailout.

REFERENCES

1. Vahanian A, Alfieri O, Andreotti F, et al. Guidelines on the management of valvular heart disease (version 2012). *Eur Heart J*. 2012;33:2451–2496. https://doi.org/10.1093/eurheartj/ehs109

2. Pineda AM, Harrison JK, Kleiman NS, et al. Incidence and outcomes of surgical bailout during TAVR. *JACC Cardiovasc Interv*. 2019;12:1751–1764. https://doi.org/10.1016/j.jcin.2019.04.026

3. Eggebrecht H, Vaquerizo B, Moris C, et al. Incidence and outcomes of emergent cardiac surgery during transfemoral transcatheter aortic valve implantation (TAVI): insights from the European Registry on Emergent Cardiac Surgery during TAVI (EuRECS-TAVI). *Eur Heart J*. 2018;39:676–684. https://doi.org/10.1093/eurheartj/ehx713

4. Compagnone M, Dall'Ara G, Grotti S, et al. Transcatheter aortic valve replacement without on-site cardiac surgery. *JACC Cardiovasc Interv*. 2023;16:3026–3030. https://doi.org/10.1016/j.jcin.2023.09.020

5. Vlastra W, Jimenez-Quevedo P, Tchétché D, et al. Predictors, incidence, and outcomes of patients undergoing transfemoral transcatheter aortic valve implantation complicated by stroke. *Circ Cardiovasc Interv*. 2019;12:e007546. https://doi.org/10.1161/CIRCINTERVENTIONS.118.007546

6. van Nieuwkerk AC, Aarts HM, Hemelrijk KI, et al. Bleeding in patients undergoing transfemoral transcatheter aortic valve replacement. *JACC Cardiovasc Interv*. 2023;16:2951–2962. https://doi.org/10.1016/j.jcin.2023.10.011

7. Kappetein AP, Head SJ, Genereux P, et al. Updated standardized endpoint definitions for transcatheter aortic valve implantation: the Valve Academic Research Consortium-2 consensus document (VARC-2). *Eur J Cardiothorac Surg*. 2012;42:S45–S60. https://doi.org/10.1093/ejcts/ezs533

8. Barbanti M, Yang T-H, Rodès Cabau J, et al. Anatomical and procedural features associated with aortic root rupture during balloon-expandable transcatheter aortic valve replacement. *Circulation*. 2013;128:244–253. https://doi.org/10.1161/CIRCULATIONAHA.113.002947

9. Pesarini G, Venturi G, Tavella D, et al. Real world performance evaluation of transcatheter aortic valve implantation. *J Clin Med*. 2021;10:1890. https://doi.org/10.3390/jcm10091890

10. Wendler O, Walther T, Schroefel H, et al. The SOURCE Registry: what is the learning curve in trans-apical aortic valve implantation? *Eur J Cardiothorac Surg*. 2011;39:853–860. https://doi.org/10.1016/j.ejcts.2010.11.018

11. van Steenbergen GJ, Olsthoorn JR, Eerdekens R, Tan E, Tonino PAL, Lam KY. Outcomes of emergent cardiac surgery after transcatheter aortic valve implantation. *Neth Heart J*. 2023;31(12):479–488. https://doi.org/10.1007/s12471-023-01820-0

12. Vahanian A, Beyersdorf F, Praz F, et al. 2021 ESC/EACTS guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43:561–632. https://doi.org/10.1093/eurheartj/ehab395

13. Hirji SA, McCarthy E, Kim D, et al. Relationship between hospital surgical aortic valve replacement volume and transcatheter aortic valve replacement outcomes. *JACC Cardiovasc Interv*. 2020;13:335–343. https://doi.org/10.1016/j.jcin.2019.09.048

14. Elbaz-Greener G, Masih S, Fang J, et al. Temporal trends and clinical consequences of wait times for transcatheter aortic valve replacement. *Circulation*. 2018;138:483–493. https://doi.org/10.1161/CIRCULATIONAHA.117.033432

15. Costa G, Barbanti M, Picci A, et al. Predictors and safety of next-day discharge in patients undergoing transfemoral transcatheter aortic valve implantation. *EuroIntervention*. 2020;16:e494–e501. https://doi.org/10.4244/EIJ-D-19-01080

16. Aarts HM, van Hemert ND, Meijs TA, et al. Percutaneous coronary intervention in patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis. *Neth Heart J*. 2023;31:489–499. https://doi.org/10.1007/s12471-023-01824-w

KEY WORDS cardiac surgery, complications, mortality, surgical bailout, transcatheter aortic valve replacement

APPENDIX For a supplemental table and a video of the interactive Central Illustration, please see the online version of this paper.